

Interview prep Session: C#

1. What is MSIL?
2. What is the compilation process in .NET?
3. What is metadata in .NET?
4. What is JIT compilation?
5. Types of JIT compilers?
6. What is CLR?
7. Responsibilities of CLR?
8. What is managed code?
9. What is unmanaged code?
10. What is an assembly?
11. What are data types in C#?
12. Categories of data types?
13. What are value types?
14. What are reference types?
15. Difference between value and reference types?
16. What is boxing?
17. What is unboxing?
18. What is var keyword?
19. What is nullable type?
20. What is default value of data types?
21. What is exception handling?
22. Keywords used in exception handling?
23. Structure of exception handling?
24. What is try block?
25. What is catch block?
26. What is finally block?
27. What is throw keyword?
28. What is custom exception?
29. What is stack trace?
30. Difference between throw and throw ex?
31. What are the four pillars of OOP?
32. What is encapsulation?
33. What is inheritance?
34. Types of inheritance in C#?
35. What is polymorphism?
36. Types of polymorphism?

37. What is method overloading?
38. What is method overriding?
39. What is abstraction?
40. What is abstract class?
41. What is interface?
42. Difference between abstract class and interface?
43. What is constructor?
44. Types of constructors?
45. What is static keyword?
46. What is sealed class?
47. What is partial class?
48. What is object class?
49. What is ToString() method?
50. What is difference between class and struct?

Scenario Based Question

1. Application Crashes Due to Unhandled Exception

Scenario: Your application crashes when dividing two numbers because the denominator sometimes becomes zero. How would you handle this?

2. Need to Restrict Direct Access to a Variable

Scenario: You have a class Employee with a salary variable that should not be directly modified outside the class.

3. Multiple Classes Need Same Method Behavior

Scenario: Classes Car and Bike both need a method Start(). How would you design this using OOP?

4. Need to Reuse Code from Existing Class

Scenario: You already have a class Employee and need a Manager class with additional features.

5. Handling Different Input Types in Same Method

Scenario: You want a method Add() to work for two integers and three integers.

6. Runtime Behavior Needs to Change

Scenario: A base class Shape has a method Draw(). Different shapes must implement their own drawing logic.

7. Value Type Needs to Be Stored in Object Variable

Scenario: You store an integer inside an object variable.

8. Object Value Needs to Be Converted Back to Integer

Scenario: You need to retrieve the integer value stored in an object.

9. Resource Must Always Be Closed

Scenario: You are reading a file and want to ensure the file is closed even if an exception occurs.

10. Need to Hide Implementation Details

Scenario: You want users to know that a class performs calculations but should not see how it works internally.